Supplementary Information

Parallel transcriptomics reveals insect developmental shifts and gall-associated host transcriptional differences in *Quadrastichus erythrinae* and *Erythrina variegata*

Serdar DİNÇ

Supplementary Figures S1–S8

Supplementary Figure S1. Insect principal component analysis

Supplementary Figure S2. Insect sample correlations

Supplementary Figure S3. Insect volcano plot

Supplementary Figure S4. Insect MA plot

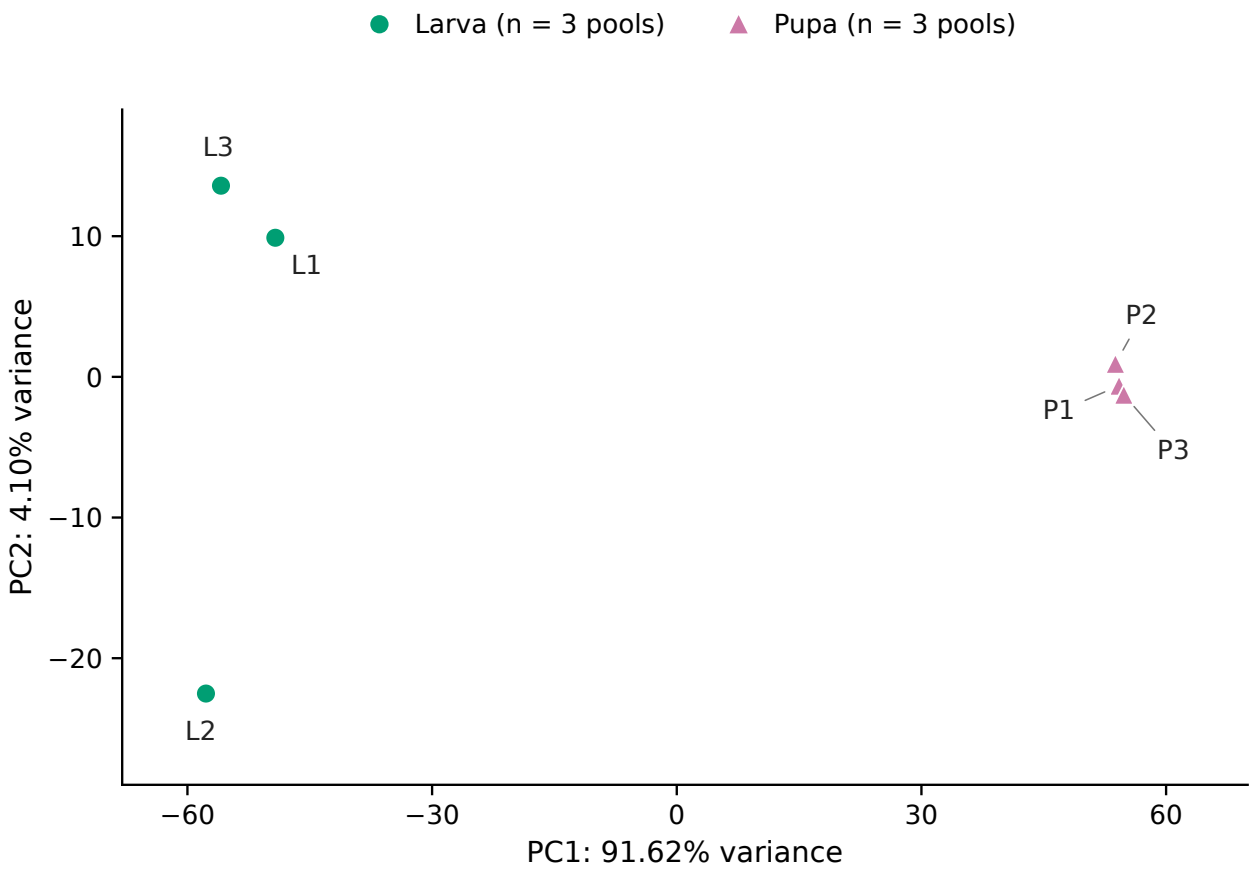
Supplementary Figure S5. Plant principal component analysis

Supplementary Figure S6. Plant sample correlations

Supplementary Figure S7. Plant volcano plot

Supplementary Figure S8. Plant MA plot

Supplementary Figure S1



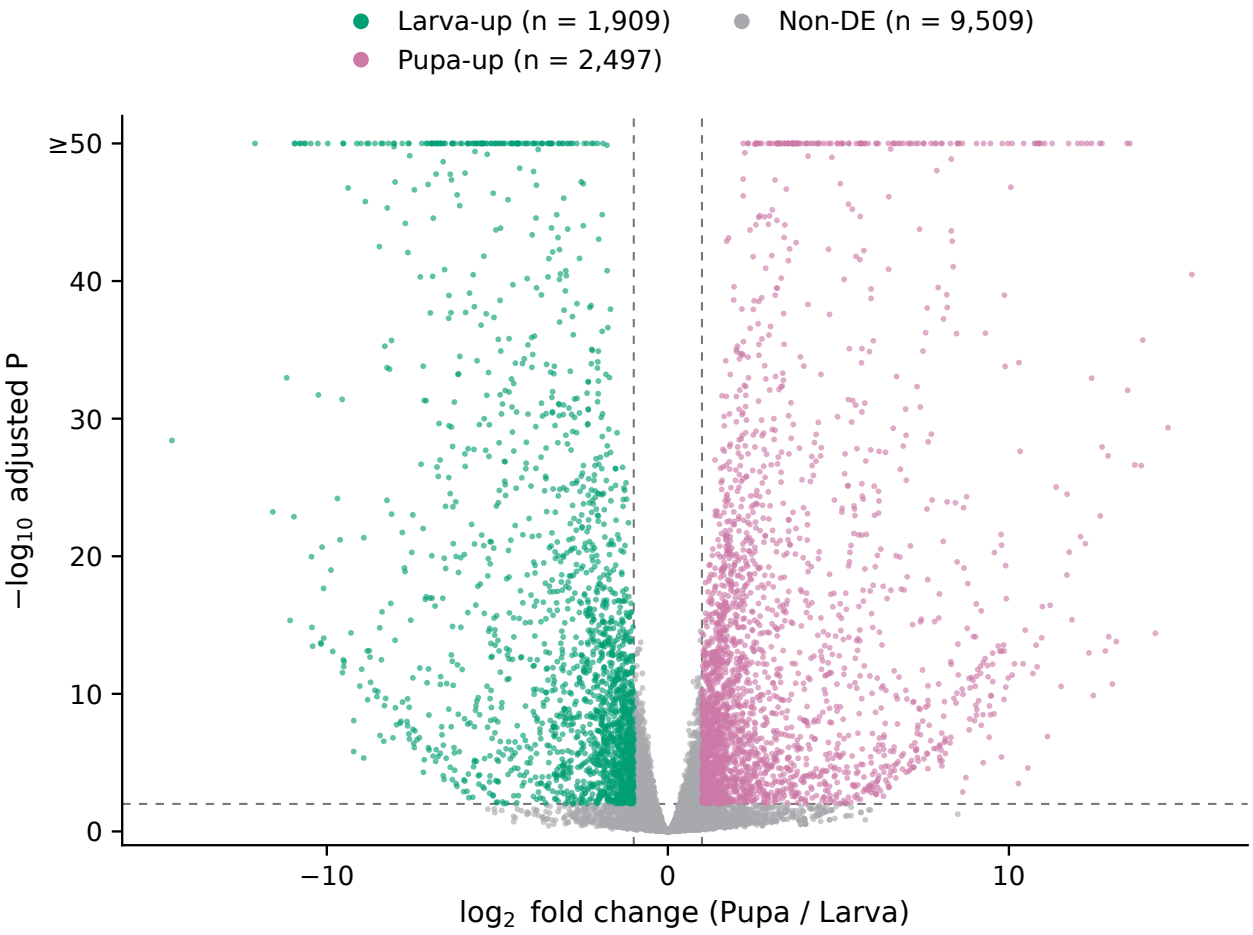
Supplementary Figure S1. Principal component analysis of variance-stabilized gene-level counts from larval and pupal *Q. erythrinae* libraries. Each labeled point represents one pool of 10 individuals (n = 3 pools per stage). The primary analysis used blind = FALSE and the 500 genes with the highest variance. PC1 and PC2 explained 91.62% and 4.10% of the variance, respectively. Larvae are shown in green and pupae in pink. The stages separated along PC1, while L2 was displaced from the other larval samples along PC2. A blind = TRUE sensitivity analysis retained the stage separation along PC1.

Supplementary Figure S2



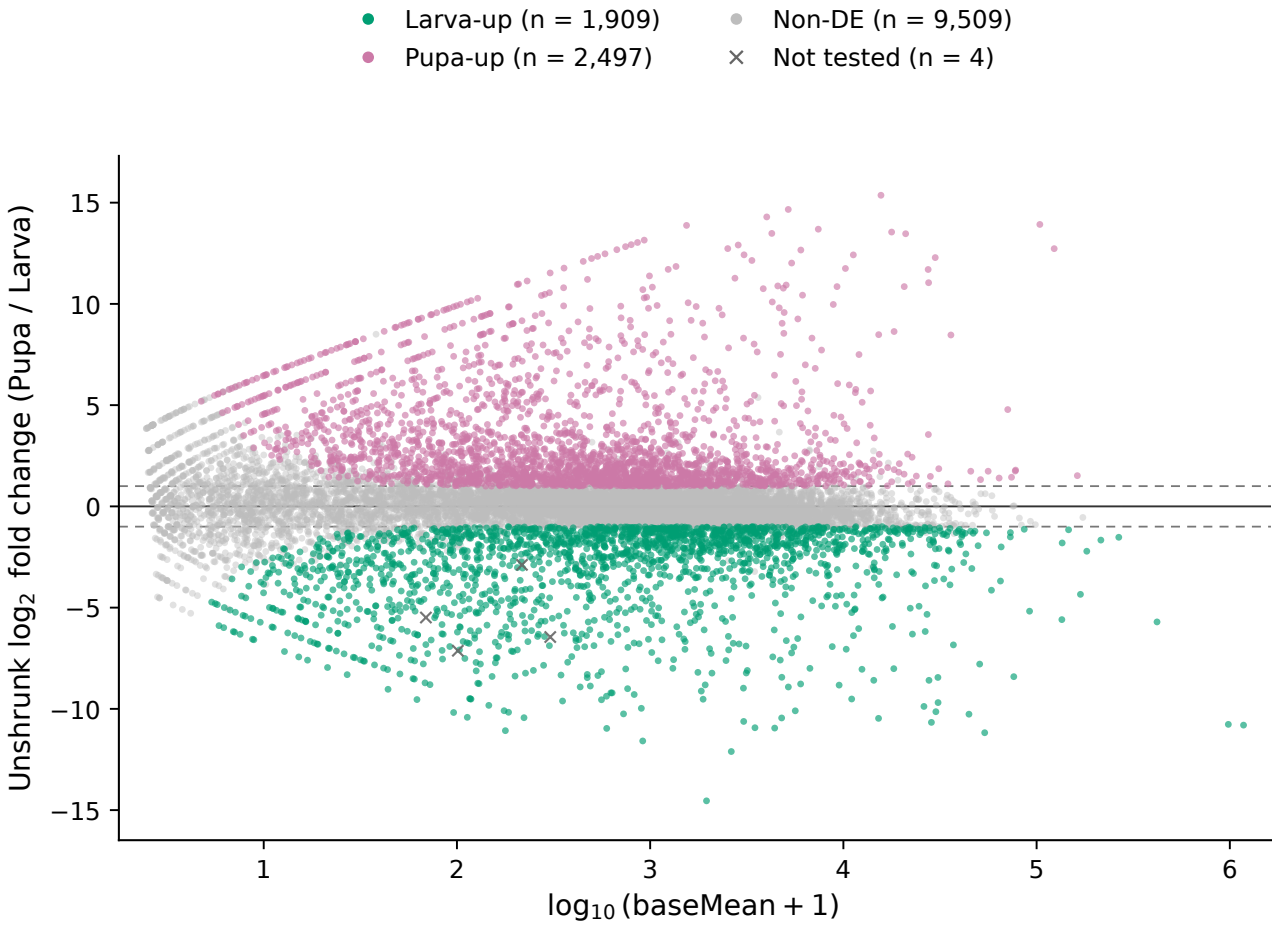
Supplementary Figure S2. Pearson correlations among larval and pupal *Q. erythrinae* libraries. Correlations were calculated from the primary variance-stabilized expression matrix (blind = FALSE) containing 13,919 retained genes. Rows and columns denote three larval pools (L1–L3) and three pupal pools (P1–P3), each containing 10 individuals. Cell values show Pearson correlation coefficients; the color scale indicates correlation strength.

Supplementary Figure S3



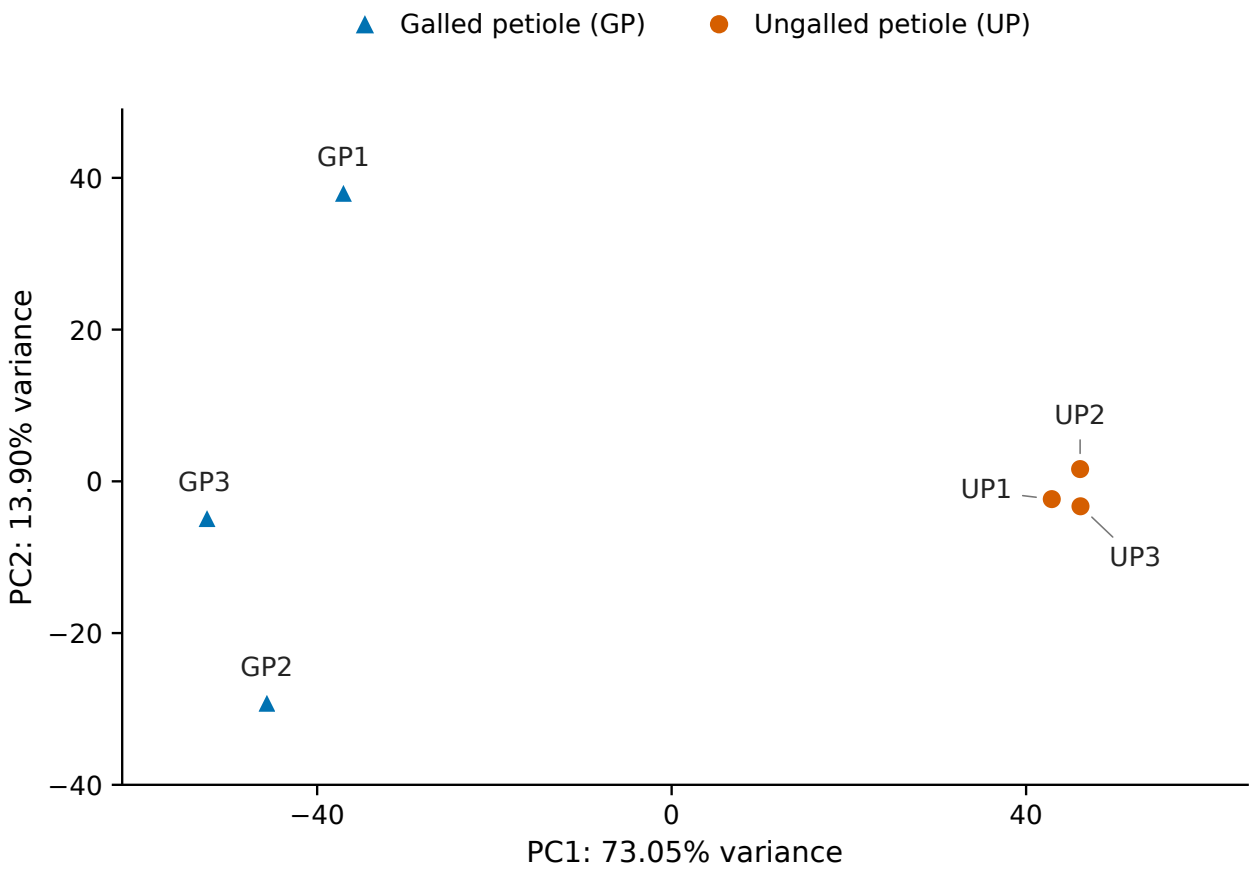
Supplementary Figure S3. Volcano plot of differential expression between pupal and larval *Q. erythrinae*. Positive unshrunk $\log_2$ fold changes indicate higher transcript abundance in pupae and negative values indicate higher abundance in larvae. Pupa-up genes are shown in pink and larva-up genes in green. DEG classification required BH-adjusted $P < 0.01$ and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1; dashed lines indicate these cutoffs. Non-DE denotes genes with an evaluable adjusted P value that did not meet both criteria. The plot includes 13,915 genes with evaluable adjusted P values. Values of -$\log_{10}$ (adjusted P) above 50 are displayed at 50; this display cap does not alter the underlying adjusted P values or DEG classification.

Supplementary Figure S4



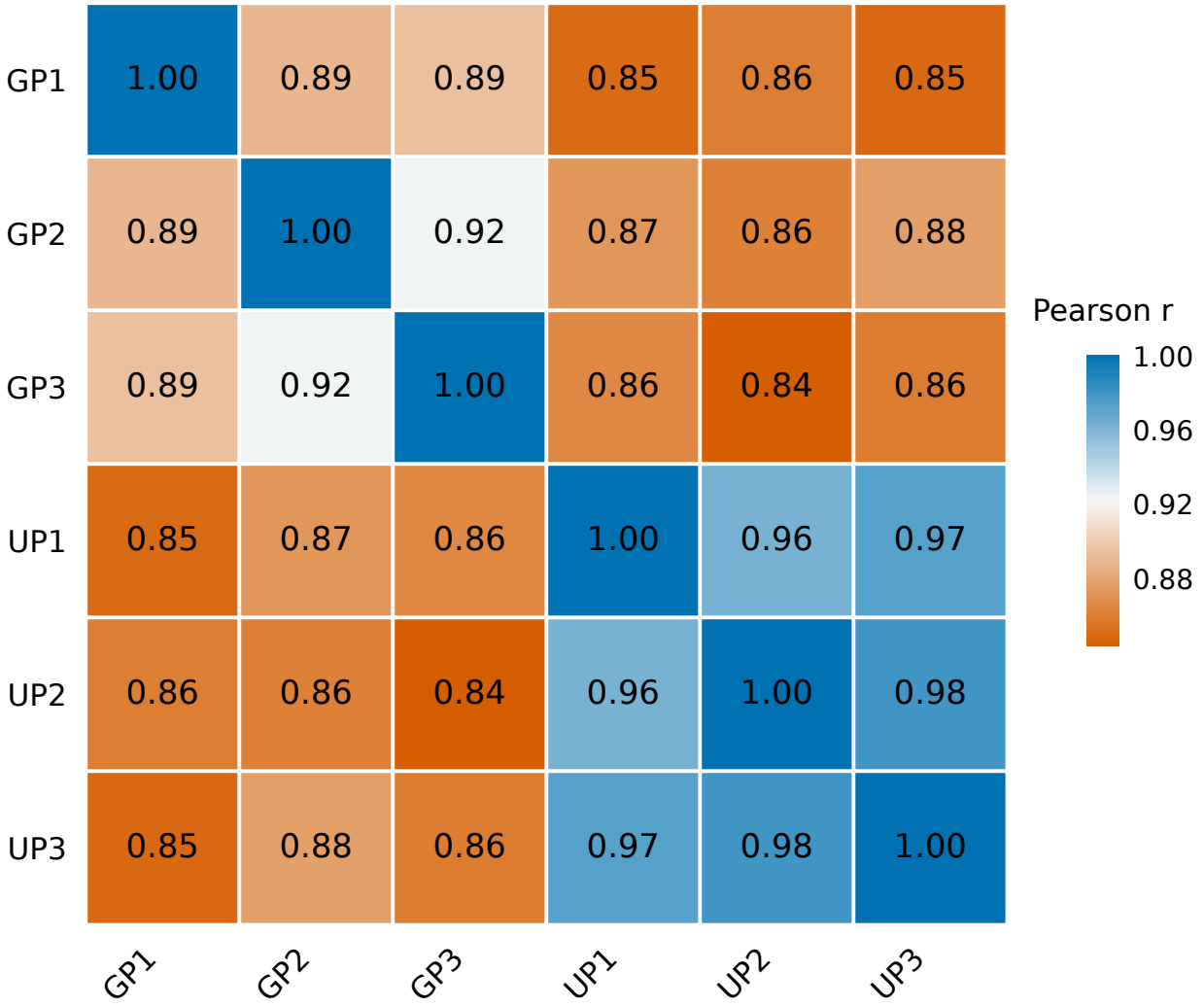
Supplementary Figure S4. MA plot of insect developmental-stage expression differences. Each point represents one of 13,919 retained genes. The x-axis shows $\log_{10}$(baseMean + 1), where baseMean is mean normalized count abundance, and the y-axis shows the unshrunk $\log_2$ fold change for pupa versus larva. Green and pink points indicate larva-up (n = 1,909) and pupa-up (n = 2,497) genes, respectively, under the primary DEG criteria of BH-adjusted $P < 0.01$ and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1. Non-DE genes (gray; n = 9,509) have an available adjusted P value but do not meet both criteria. Not tested genes (crosses; n = 4) have an unavailable adjusted P value and are shown at their actual finite fold-change estimates. The solid line marks zero and dashed lines mark $\log_2$ fold changes of -1 and 1. Legend counts are gene counts.

Supplementary Figure S5



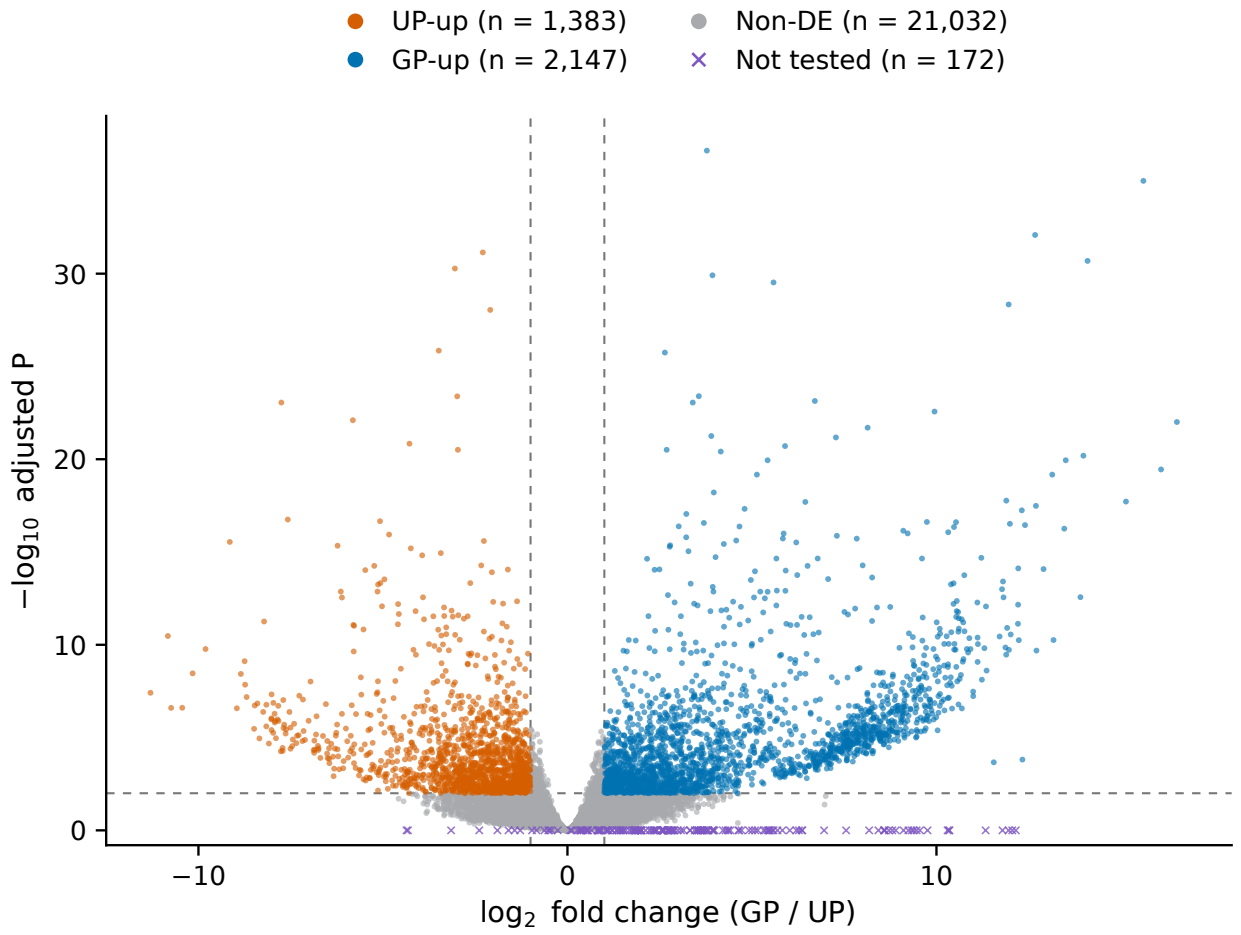
Supplementary Figure S5. Principal component analysis of the final refined-mask *E. variegata* expression dataset. Each point represents one whole-petiole library, with three galled petioles (GP, blue triangles) and three ungalled petioles (UP, orange circles) collected from the same host tree during one sampling occasion (n = 3 petioles per condition). PCA was based on variance-stabilized counts. PC1 and PC2 explained 73.05% and 13.90% of the variance, respectively. GP and UP samples separated along PC1; GP samples showed a wider spread than UP samples, particularly along PC2.

Supplementary Figure S6



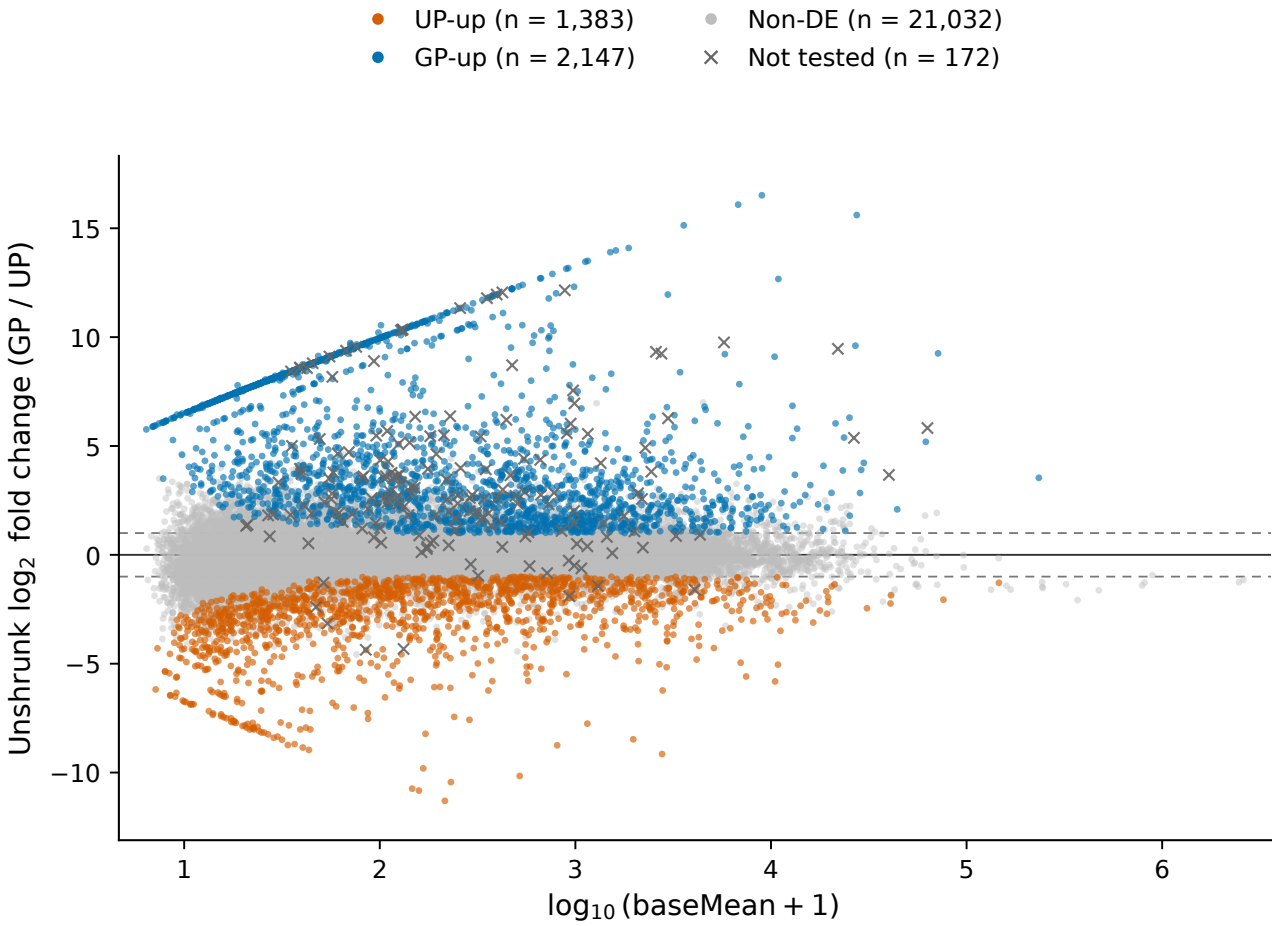
Supplementary Figure S6. Pearson correlations among *E. variegata* whole-petiole libraries. Correlations were calculated from the variance-stabilized final refined-mask expression matrix (blind = TRUE) containing 24,734 retained genes. Rows and columns denote three galled petioles (GP1–GP3) and three ungalled petioles (UP1–UP3) collected from one host tree during one sampling occasion. Cell values show Pearson correlation coefficients; the color scale indicates correlation strength.

Supplementary Figure S7



Supplementary Figure S7. Differential gene expression between galled and ungalled *E. variegata* petioles. The volcano plot shows the final refined-mask GP-versus-UP comparison using unshrunk DESeq2 coefficients. Positive $\log_2$ fold changes indicate higher transcript abundance in GP and negative values indicate higher abundance in UP. GP-up genes are blue and UP-up genes are orange. DEG classification required BH-adjusted $P < 0.01$ and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1; dashed lines indicate these cutoffs. Non-DE denotes genes with an evaluable adjusted P value that did not meet both criteria. Not tested denotes 172 genes lacking an evaluable adjusted P value; crosses at y = 0 show their display positions only and do not represent imputed P values. Large absolute unshrunk coefficients should be interpreted alongside mean normalized counts and coefficient standard errors, provided in the source data.

Supplementary Figure S8



Supplementary Figure S8. MA plot of gall-associated plant expression differences. Each point represents one of 24,734 retained genes after the 16,404-gene sequence-provenance mask and frozen count filter. The x-axis shows $\log_{10}$(baseMean + 1), where baseMean is mean normalized count abundance, and the y-axis shows the unshrunk $\log_2$ fold change for galled (GP) versus ungalled (UP) whole petioles. Blue and orange points indicate GP-up (n = 2,147) and UP-up (n = 1,383) genes, respectively, under the primary DEG criteria of BH-adjusted P < 0.01 and an absolute estimated, unshrunk $\log_2$ fold change ≥ 1. Non-DE genes (gray; n = 21,032) have an available adjusted P value but do not meet both criteria. Not tested genes (crosses; n = 172) have an unavailable adjusted P value and are shown at their actual finite fold-change estimates. The solid line marks zero and dashed lines mark $\log_2$ fold changes of -1 and 1. Legend counts are gene counts.